Sample Public School

Unit Test: Electricity

Subject: Physics Class: Class 10

Time allowed: 45 minutes **Maximum marks: 20**

Name: _____ Roll no.: _____

Class & section: _____ Date: _____

General instructions

1. All questions are compulsory.
2. Section A has 2 multiple choice questions of 1 mark each.
3. Show the formula and working for every numerical question.

SECTION A

Multiple Choice Questions

Choose the correct option. (2 × 1 = 2 marks)

1. The SI unit of electric resistance is: [1]

A) volt B) ampere

C) ohm D) watt

2. Which instrument is used to measure electric current in a circuit? [1]

A) voltmeter B) ammeter

C) galvanometer D) potentiometer

SECTION B

Short Answer Questions

Answer in two or three sentences. (2 × 2 = 4 marks)

3. State Ohm's law. [2]

4. Define electric power and give its SI unit. [2]

SECTION C

Numerical Questions

Show the formula, working and units. (2 × 3 = 6 marks)

5. A 12 V battery is connected across a 4 Ω resistor. Find the current in the circuit and the power used by the resistor. [3]

6. Three resistors of 2 Ω, 3 Ω and 6 Ω are connected (a) in series and (b) in parallel. Find the equivalent resistance in each case. [3]

SECTION D

Long Answer Questions

Answer in detail. (2 × 4 = 8 marks)

7. Why are household appliances connected in parallel and not in series? Give three reasons. [4]

8. Explain the heating effect of electric current. State Joule's law of heating and give two applications. [4]

— End of question paper —